

TERRY UWAGBAE OKO-ODION

terryokoodion@gmail.com

<https://terryoko.github.io>

RESEARCH INTEREST

AI Evaluation

Clinical AI Safety

Reliable and Trustworthy NLP

EDUCATION

BSc (Honours) Ambrose Alli University, Computer Science July 2024
Thesis: Designed and implemented a web-based academic repository,
focusing on data integrity

GRE 337/340 (Quant: 170, Verbal: 167)

RESEARCH EXPERIENCE

Machine Learning Collective (MLC) October 2025 – Present

Independent AI/ML Researcher

- Developing shift-aware safety metrics (Risk-Coverage Shift, AURC) to audit confidence calibration in clinical LLMs when transferred from Western (USMLE) to African (AfriMed) contexts (geographic shift).
- Co-leading an audit of the XNLI benchmark for 11 African languages, engineering a data pipeline to scale test sets by 800% (from 600 to 5,010 samples per language) to ensure statistical reliability in model evaluation
- Investigating evaluation brittleness in LLMs, quantifying how small-sample benchmarks mask performance gaps between English and low-resource African languages
- Benchmarking the brittleness of standard instruction tuning in small open-weight models (Llama-3.2-3B, Gemma-3-4B) across Hausa, Yoruba, and Swahili, specifically quantifying accuracy degradation caused by orthographic mismatches (e.g., Ajami vs. Latin scripts)
- Developing and validating Native-Label Self-Translation Prompting (NL-STP), a novel inference strategy that forces intermediate reasoning in the target language to decouple prompt design failures from latent model capability

AI for Food Allergies (Hugging Face Science) September 2025 – Present

Research Assistant

- Engineered a balanced distribution strategy for the Goldberg microbial signature dataset (700+ samples) to prevent the model from learning spurious temporal correlations
- Restructured clinical metadata for 233 individuals by resolving ambiguous SRS IDs and inconsistent temporal labels, ensuring causal integrity for the binary classification of IgE-mediated food allergies
- Collaborating with domain experts to define clinically rigorous evaluation metrics that prioritize worst-case safety over average-case accuracy in high-stakes healthcare diagnostics

Honours Thesis

- Designed and implemented a web-based academic repository, focusing on data integrity as the foundational layer for trustworthy AI
- Engineered a structured metadata schema and validation pipelines to ensure data consistency and combat incompleteness bias
- Implemented access controls and audit trails to establish clear data provenance, a critical requirement for accountable AI systems

PUBLICATIONS, WORKSHOPS & MANUSCRIPTS

5. **Oko-odion, T. U.**, Tiwari, A., Amin, E., Asama, T. & Alele, G. (2026). "Selective Risk Inflation under Geographic Distribution Shift in Small Medical LLMs." **[GEM, ACL 2026 (in review)]**
4. Tiwari, A., **Oko-odion, T. U.**, Nwokocha, H., Ogunremu, O. & Egbewale, J. (2026). "From Script to Semantics: Prompting Strategies for African NLI." **[RAIL, LREC 2026 (in review)]**
3. Tiwari, A., Ogunremu, O., **Oko-odion, T. U.**, Nwokocha, H. & Egbewale, J. (2026). "Sample-Size Scaling of the African Languages NLI Evaluation." [\[AfricaNLP, EACL 2026\]](#)
2. Okereke, R. O., Alele, G. O., Azeez, O. L., **Oko-odion, T. U.**, Samson, I. O., Anosike, C. N., Owan, F. O. & Nnamani, C. C. (2024). "The Cloud Security Revolution: Unlocking the Potential of AI and Machine Learning to Stay Ahead of Threats." Asian Journal of Science, Technology, Engineering, and Art, 2(5), 735-743
1. **Oko-odion, T. U.**, Ruth, O. O., Chijioke, N. A., Chinenye, C. N., Fakokunde, B. D., Olamide, A. & Oghenerukevwe, S. I. (2024). "Data Privacy Defense: Strategies to Mitigate Internal and External Hacking Risks." IJAEM, 6(9), 417-426

TEACHING EXPERIENCE

Instructor's Assistant - CSC 101 Introduction to Computer Science

Jan. 2021 - Nov. 2021

Department of Computer Science, Ambrose Alli University

- Conducted lectures, held discussions, and provided guidance to students in CSC 101
- Facilitated active learning experiences, encouraging student participation
- Challenged and motivated students through in-depth one-on-one or group discussions
- Assessed student learning through online polls, assignments and quizzes

Instructor's Assistant - CSC 305 Artificial Intelligence

June 2023 - Jan. 2024

Department of Computer Science, Ambrose Alli University

- Collaborated with faculty to evaluate student projects, ensuring consistent grading of AI concepts and Python programming assignments
- Graded assignments and mentored students in programming fundamentals
- Acted as a link between students and faculty, addressing any concerns or issues raised by the students and facilitating effective communication

TECHNICAL SKILLS

Research Computing: Python (PyTorch, TensorFlow, NumPy, Pandas and Scikit-learn), Statistical Analysis & Data Visualization, Git Version Control and Docker Containerization

Machine Learning Research: Deep Learning, Model Evaluation & Benchmark Design, Data Preprocessing & Feature Engineering, Experimental Methodology and Statistical Testing

Software Engineering: Ruby on Rails, FastAPI and Web-based Repository Development

MENTORING ACTIVITIES

Mentored junior undergraduates in Python programming and in managing the community		
Will Osezele, Google Developer Student Club AAU		2023-Present
Destiny Egbon, Google Developer Student Club AAU		2023-Present
Chiamaka Esther, Google Developer Student Club AAU,		2023-Present
Emmanuel Ogbebor, Google Developer Student Club AAU,		2023-Present

SERVICE

Co-organizer, Django Girls Bootcamp AAU	2024
Core Team Member & Mentor, Google Developer Student Club AAU	2023-2024
Volunteer, TEDx AAU Ekpoma	2023
Director of Information, National Association of Physical Science Students	2022-2023

SCHOLARSHIP OFFERS & HONORS (TOTAL VALUE: ~\$57,030)

Graduate Merit Scholarship (Emory University), 2026: Awarded a 20% tuition fellowship for each term until graduation for the MS in Computer Science, valued at approximately \$18,500 total (Declined).

Graduate Dearborn Scholarship (University of Michigan), 2025: Awarded up to \$10,000 annually based on outstanding academic qualifications (Declined).

Strategic Graduate Scholarship (Stevens Institute of Technology), 2025: Awarded a \$11,000 merit-based scholarship (\$7,000 for first two semesters, \$4,000 for the third) for highly qualified prospective students (Declined).

Dean’s Scholarship (Northeastern University), 2026: Awarded \$6,000 tuition scholarship for MS in Artificial Intelligence (Declined).

INGRYD/Lagos State Employment Trust Fund (LSETF) Scholar, 2024: Awarded a \$1,500 full-tuition scholarship for advanced technical training in Tech.

Matthew Okpebholo Foundation Annual Scholarship, 2023: Awarded for academic excellence and leadership.

GRANTS (TOTAL SECURED: \$1,000)

Dataset Validation Grant, 2025: Awarded \$1,000 for data collection and validation for the "From Script to Semantics: Prompting Strategies for African NLI" research paper currently under review at LREC (RAIL Workshop)

PROFESSIONAL AFFILIATIONS

Machine Learning Collective (MLC)	2025-Present
Member, Black in AI	2024-Present
Google Developer Student Club	2023-2024

PRESENTATIONS & TRAINING

Presentation

- Honours Thesis Defense, "Design of a Web-Based Academic Repository for Data Integrity," Ambrose Alli University, 2023
- Seminar Presentation, "Principles of Zero Trust in Cyber Security Focusing on Corporate Setting," Ambrose Alli University, 2023

Training

- Invited Speaker, "Cybersecurity Awareness Month," Google Developer Student Club, 2023
- Training Facilitator, "Hacktoberfest: Version Control with Git & GitHub," 2023

LANGUAGES

English: Native Language, Distinguished levels in Listening, Speaking, Reading, and Writing